Parul University
Faculty of Engineering & Technology
Department of Applied Sciences and Humanities
1st Year B. Tech Programme (CSE(All Branches), IT) Semester-2
Linear Algebra (03019102BS01)
Unit–1 Matrices (Tutorial-1) 2025-26

Q-1. Identify if the following matrices are Symmetric, skew symmetric or not.

(a) $\begin{bmatrix} 0 & 2 & 3 \\ -2 & 0 & -4 \\ -3 & 4 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}$

Q-2. Identify if the following matrices are orthogonal or not.

(a) $\begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix}$ (b) $\frac{1}{7}\begin{bmatrix} 3 & 2 & 6 \\ -6 & 3 & 2 \\ 2 & 6 & -3 \end{bmatrix}$

Q-3. Which of the following matrices are in row-echelon form, reduced-row echelon form, or both? Justify your answer with appropriate reasons.

(a) $A = \begin{bmatrix} 2 & 2 & -1 \\ 5 & -1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$ (b) $B = \begin{bmatrix} 2 & 3 & -5 \\ 0 & 1 & -3 \\ -1 & -4 & 3 \end{bmatrix}$ (c) $C = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 2 & -3 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

Q-4. Find Rank of the following matrices:

(a) A= $\begin{bmatrix} 1 & 5 & 4 \\ 0 & 3 & 2 \\ 2 & 13 & 10 \end{bmatrix}$ (b) B = $\begin{bmatrix} 1 & 3 & 2 & 1 \\ 2 & 3 & 3 & 2 \\ 3 & 4 & -1 & 3 \\ 6 & 10 & 4 & 6 \end{bmatrix}$ (c) C = $\begin{bmatrix} 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \\ 4 & 5 & 6 & 7 \\ 9 & 10 & 11 & 12 \end{bmatrix}$ (d) $D = \begin{bmatrix} 1 & 2 & -3 & 1 \\ -1 & 0 & 3 & 4 \\ 0 & 1 & 2 & -1 \\ 2 & 3 & 0 & -3 \end{bmatrix}$

Q-5. Find the value of λ for which rank of the matrix $A = \begin{bmatrix} 3 & 1 & 2 \\ -1 & 4 & 5 \\ 7 & 2 & \lambda \end{bmatrix}$

(i) is less than 3 (ii) equal to 3

Q-6. For what value of x, will the matrix $A = \begin{bmatrix} 3-x & 2 & 2 \\ 1 & 4-x & 0 \\ -2 & -4 & 1-x \end{bmatrix}$ be of rank

(i) equal to 3 (ii) less than 3

Q-7. Solve the following systems of equations using Gauss elimination method.

(a) $x - 2y + 2z = 3; 2x + y + 2z = 4; 6x + 2y - 2z = 4$
(b) $2x - y + z = 3; 3x - y + 2z = 6; -5x + 8y - 4z = 2$
(c) $x + 2y + z = 5; -x - y + z = 2; y + 3z = 1$
(d)) $2x + y + 3z = 0; x + 2y = 0; y + z = 0$

Q-8. Solve the following systems of equations using LU-Decomposition method.

(a) $x_1 + x_2 + x_3 = 3; 2x_1 - x_2 - x_3 = 3; x_1 - x_2 + x_3 = 9$
(b) $2x + 3y - z = 5; 3x + 2y + z = 10; x - 5y + 3z = 0$
(c)$x + 2y + 3z = 9; 4x + 5y + 6z = 24; 3x + y - 2z = 4$

Q-9. Investigate for what values of λ and μ, the system of simultaneous equations:

$x + y + z = 6;\ x + 2y + 3z = 10; x + 2y + \lambda z = \mu$

have (i) no solution, (ii) a unique solution, and (iii) infinite number of solutions

Q-10. Find the Eigen values and Eigen vector of the following matrices. Also determine algebraic multiplicity and geometric multiplicity of the matrices wherever possible.

(a) A= $\begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$ (b) B = $\begin{bmatrix} -3 & -7 & -5 \\ 2 & 4 & 3 \\ 1 & 2 & 2 \end{bmatrix}$ (c) C = $\begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$ (d) A = $\begin{bmatrix} 3 & 0 & 0 \\ 8 & 4 & 0 \\ 6 & 2 & 5 \end{bmatrix}$

Q-11. Verify Cayley –Hamilton theorem for the matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$. Find $A^3 - 5A^2$.

Q-12 Verify Cayley –Hamilton theorem for the matrix A= $\begin{bmatrix} 1 & 2 & -2 \\ 2 & 5 & -4 \\ 2 & 7 & -5 \end{bmatrix}$ and hence find A^{-1}.

Q-13. Verify Cayley –Hamilton theorem for the matrix A= $\begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}$ and use it to find A^4 and A^3.

Q-14. Find a matrix that diagonalizes A, and determine $P^{-1}AP$ for

(i)A = $\begin{bmatrix} 2 & 2 & -1 \\ 5 & -1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$ (ii) A = $\begin{bmatrix} 2 & 3 & -5 \\ 0 & 1 & -3 \\ -1 & -4 & 3 \end{bmatrix}$

Q-15. Find PCA for the matrix $X = \begin{bmatrix} 4 & 11 \\ 8 & 4 \\ 13 & 5 \\ 7 & 14 \end{bmatrix}$